

LIS at BAREC Shared Task 2025: Multi-Scale Curriculum Learning for Arabic Sentence-Level Readability Assessment Using Pre-trained Language Models

Anya Amel Nait djoudi, Patrice Bellot, Adrian-Gabriel Chifu

Aix-Marseille Université, CNRS, LIS

{anya-amel.NAIT-DJOUDI, patrice.bellot, adrian.chifu}@univ-amu.fr

Abstract

Sentence-level readability assessment, which measures how easily individual sentences can be understood, has seen significant advances in English. However, Arabic readability assessment remains underexplored, primarily due to the language’s morphological complexity and the scarcity of fine-grained annotated datasets. To address this gap, we leveraged the BAREC corpus, which provides 69K sentences annotated across 19 readability levels, enabling us to develop and compare five different modeling strategies ranging from lightweight classifiers to fine-tuned Arabic language models. Our experiments revealed that task-specific pretraining with CamelBERT yielded substantial performance gains, while curriculum learning offered benefits in specific scenarios. Ultimately, direct fine-tuning achieved state-of-the-art performance (QWK = 82.4). Through detailed error analysis, we identified that models struggled most with distinguishing between the lower readability level 2 and higher readability levels (15-19), highlighting the inherent challenges in fine-grained Arabic readability modeling across the full spectrum of proficiency levels.

1 Introduction

Readability assessment measures how easily a text can be read and understood by its target audience. It is crucial in various contexts including pedagogical settings, foreign language learning, health literacy (Djoudi et al., 2025), and content accessibility (Xia et al., 2016; Vajjala and Meurers, 2012; Collins-Thompson and Callan, 2004; Fox and Dugan, 2013). To enable automatic assessment, many resources have been made available ranging from datasets with annotated readability levels to readability assessment models. While progress in English readability assessment has been extensive (Azpiazu and Pera, 2019; Deutsch et al., 2020; Qiu et al., 2021; Devlin et al., 2019), Arabic readability assessment remains less studied. Arabic presents

unique challenges due to its morphological richness, complex derivational patterns, and limited availability of fine-grained annotated resources. The available datasets predominantly employ binary (Soliman and Familiar, 2024) or ternary classification schemes (Al-Khalifa and Al-Ajlan, 2010), which train models with an oversimplified view of reading proficiency. Fine-grained readability levels offer the potential to better capture the continuous spectrum of literacy levels across diverse readers and provide more nuanced assessments that align with real-world reading abilities.

To address these limitations, we utilize the newly introduced BAREC corpus (Balanced Arabic Readability Evaluation Corpus)¹ (Elmadani et al., 2025b), a large-scale, fine-grained corpus containing over 69,000 sentences from 1,922 documents. The corpus spans 19 readability levels, from kindergarten (1) to postgraduate (19), covering diverse genres and domains (Arts & Humanities, Social Sciences, STEM) across three readership groups (Foundational, Advanced, Specialized).

Our contributions include systematic evaluation of five distinct model architectures, spanning lightweight MLP classifiers over pre-trained embeddings to full progressive and direct fine-tuning of Arabic language models. We demonstrate that:

1. task-specific pretraining is essential, with readability-focused CamelBERT substantially outperforming general-purpose models;
2. curriculum learning provides situational benefits in fine-tuning settings;
3. direct fine-tuning achieves state-of-the-art performance (QWK = 82.4);
4. comprehensive error analysis reveals difficulty in distinguishing lower readability level 2 and higher readability levels (15 to 19).

¹BAREC corpus: <https://huggingface.co/datasets/CAMEL-Lab/BAREC-Shared-Task-2025-sent>

2 Background

Text difficulty evaluation traditionally used surface-level formulas like DCRS (Dale and Chall, 1948), FKGL (Kincaid et al., 1975), Dawood and El-Heeti (Al-Dawsari, 2004), AARI (Al Tamimi et al., 2014), and OSMAN (El-Haj and Rayson, 2016). The development of readability corpora enabled richer statistical and neural modeling approaches.

These corpus-driven advances have been most prominent in English, with resources including WeeBit (Vajjala and Meurers, 2012), Newsela (Xu et al., 2015), Cambridge (Xia et al., 2016), OneStopEnglish (Vajjala and Lučić, 2018) (document-level), S1131 (Štajner et al., 2017), CEFR-SP (Arase et al., 2022) (sentence-level), and cross-lingual corpora MDTE (De Clercq and Hoste, 2016), CompDS (Brunato et al., 2018). For Arabic, efforts such as (Hazim et al., 2022) have facilitated Arabic readability annotations, leading to the development of resources spanning multiple granularities: documents (Arability (Al-Khalifa and Al-Ajlan, 2010), DLI (Forsyth, 2014), Taha/Arabi21 (Taha-Thomure, 2017), ZAEBC (Habash and Palfreyman, 2022), QAES (Bashendy et al., 2024)), sentences (README++ (Naous et al., 2024), DARES (El-Haj et al., 2024), BAREC (Elmadani et al., 2025b)), and words (KELLY (Kilgariff et al., 2014), SAMER (Al-Khalil et al., 2020), Arabic Vocab Profile (Soliman and Familiar, 2024), extended SAMER (Alhafni et al., 2024)).

Building on these corpus development efforts, broader research has focused on Arabic readability modeling using diverse strategies (Liberato et al., 2024). To advance this field further and provide a standardized evaluation framework, the BAREC Shared Task 2025 (Elmadani et al., 2025a) introduces 19-level fine-grained readability prediction with three tracks: Strict (BAREC corpus only), Constrained (BAREC & SAMER corpora), and Open (any public data). We participate in the Strict Track for sentence-level assessment, predicting Arabic sentence difficulty on a 19-point scale (1 = easiest, 19 = hardest) using models trained exclusively on BAREC training data.

3 System Overview

We experiment with CAMELBERTMix_MLP and CAMELBERTWCE_MLP, which combine contextual embeddings from Arabic BERT models with a lightweight multilayer perceptron (MLP) classifier. CAMELBERTMix_MLP employs bert-base-

arabic-camelbert-mix ² (Inoue et al., 2021), a general-purpose encoder trained on a mix of modern standard, dialectal, and classical Arabic, while CAMELBERTWCE_MLP uses readability-camelbert-word-CE ³ (Elmadani et al., 2025b), fine-tuned on the same dataset as this shared task. Sentences are tokenized with a maximum length of 256 tokens, encoded, and aggregated via mean pooling over attention-masked hidden states to yield 768 dimensional embeddings. These embeddings are normalized using RobustScaler and fed into the MLP, which was selected via 5-fold stratified cross-validation. The best-performing configuration is a two-layer architecture (256-128 neurons), ReLU activation, L2 regularization ($\alpha = 0.01$), trained with Adam optimization (initial learning rate 10^{-4} , adaptive scheduling), batch size 64, and a maximum of 300 epochs.

P_CAMELBERTWCE_MLP architecture implements a progressive multilayer perceptron (MLP) trained with a curriculum learning strategy. The model is trained sequentially through multiple stages of increasing granularity (3-5-7-19 levels), utilizing the same 768 dimensional embeddings as CAMELBERTWCE_MLP. The 19-level ground truth labels are collapsed into intermediate targets using custom binning strategies to create more balanced distributions: 3-level bins [0, 7, 13, 19], 5-level bins [0, 4, 8, 12, 16, 19], and 7-level bins [0, 3, 6, 9, 12, 15, 17, 19]. Training begins with a simple 3-level classifier (2-layer architecture with 256 and 128 neurons), then progresses to 5-level and 7-level classifiers, and finishes with a 19-level classifier (3-layer architecture with 512, 256, and 128 neurons). The stage-specific architecture scales with task complexity. A weight transfer mechanism preserves learned hidden layer representations between stages. The model is optimized using Adam with an adaptive learning rate (initial $\eta = 10^{-3}$, reduced to 5×10^{-4} during transfer phases), a batch size of 64, and L2 regularization ($\alpha \in [0.008, 0.01]$) and a maximum of 100-300 epochs per stage depending on complexity.

PFT_CAMELBERTWCE implements a Progressive CamelBERT (Inoue et al., 2021) Fine-tuning approach that fine-tunes the CAMEL-Lab/readability-camelbert-word-CE transformer model through curriculum learning stages [3, 5, 7, 19]. The ap-

²<https://huggingface.co/CAMEL-Lab/bert-base-arabic-camelbert-mix>

³<https://huggingface.co/CAMEL-Lab/readability-camelbert-word-CE>

proach learns dynamic label mappings from dataset annotations rather than using fixed binning strategies like we did in P_CAMELBERTWCE_MLP. Training begins with 3-level classification using a dropout-regularized classification head (dropout=0.3), progressively transferring the fine-tuned BERT encoder weights to subsequent stages while initializing fresh classification heads for each target granularity. Training uses AdamW optimization with linear warmup scheduling, adaptive learning rates (2e-5 initial, 1e-5 for transfer stages). The intuition is that learning coarse readability distinctions (3-5-7 levels) first provides foundational representations that will improve fine-grained (19 levels) classification performance.

FT_CAMELBERTWCE serves as an ablation study to test the core hypothesis behind PFT_CAMELBERTWCE. It eliminates the progressive curriculum learning stages to perform direct, end-to-end fine-tuning of the CAMEL-Lab/readability-camelbert-word-CE model on the full 19-level classification task. The motivation for this simpler approach is twofold. First, it questions whether a powerful pre-trained transformer inherently possesses the latent linguistic understanding to discern fine-grained readability distinctions without being guided through coarser labels. Second, it tests if the considerable computational and architectural overhead of multi-stage progressive training is justified, or if a single-stage model can achieve comparable performance more efficiently. To ensure a fair comparison, FT_CAMELBERTWCE retains the same optimization strategy (AdamW, linear warmup) and regularization (dropout=0.3) as the final stage of PFT_CAMELBERTWCE. Thus allowing us to directly attribute any performance differences to the presence or absence of the curriculum learning framework, rather than other hyperparameters.

4 Experimental Setup:

4.1 Dataset

We used the Balanced Arabic Readability Evaluation Corpus (BAREC)⁴, a large-scale, fine-grained corpus containing over 69,000 sentences from 1,922 documents. As illustrated in Table 1, the corpus spans 19 readability levels, from kindergarten (1) to postgraduate (19), which can also be collapsed into coarser 7, 5, or 3 readability levels. For more details on sentence readability annotation, re-

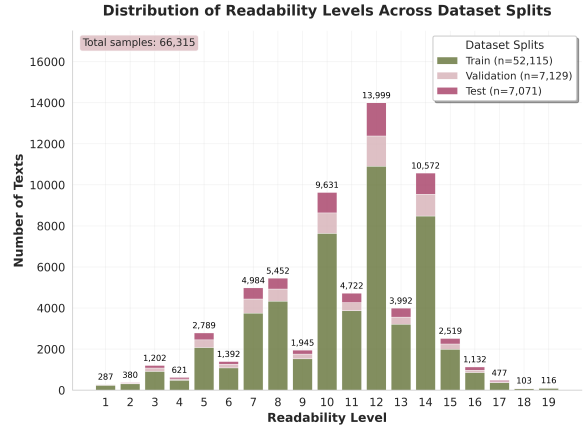


Figure 1: Distribution of the train/validation/test split in the BAREC corpus by level of readability

fer to the BAREC readability annotation guidelines by (Habash et al., 2025).

Level	Arabic	Translation	Reasoning
1	'نعم'	"Yes"	Simple single-word
19	'أو صورة مثلت في النفس من أمني'	"Or an image represented in the soul from my hope."	Complex poetic expression, rich in imagery

Table 1: Comparison of linguistic complexity between beginner and advanced Arabic expressions.

4.2 Preprocessing

Before training, we applied a multi-stage cleaning pipeline to ensure text consistency and quality. This involved removing sentences with missing values or placeholders (e.g., #NAME?), collapsing excessive whitespace, and dropping exact duplicates. For Arabic processing, we utilized CAMEL Tools (Obeid et al., 2020), a comprehensive suite of NLP resources for morphological analysis, disambiguation, dialect identification, normalization, and tokenization⁵. A key step was dediaccritization to remove short vowels and phonetic marks which are infrequent in modern Arabic and can introduce noise in NLP tasks. Dediaccritization allowed us to focus on underlying lexical forms (e.g., كِتَابٌ جَمِيلٌ was converted to كِتَاب جَمِيل, or "A beautiful book"). The pipeline removed 2.5-5% of rows, retaining 95-97% of the dataset across splits (52k train, 7.1k validation, 7k test). The final distribution of these readability levels is shown in Figure 1.

⁴BAREC corpus: <https://huggingface.co/datasets/CAMEL-Lab/BAREC-Shared-Task-2025-sent>

⁵CAMEL Tools: https://github.com/CAMEL-Lab/camel_tools

4.3 Metrics

The BAREC shared task ⁶ organizers define the readability prediction problem as an ordinal classification task and adopt the following evaluation ⁷ measures: **Quadratic Weighted Kappa (QWK) (Cohen, 1968)**: The primary evaluation metric for the shared task. It’s an extension of Cohen’s Kappa that measures agreement between predicted and reference labels, applying a quadratic penalty to larger misclassifications. **Accuracy (Acc)**: The percentage of exact matches between predicted and gold labels on the 19-level scale Acc^{19} . Variants Acc^7 , Acc^5 , and Acc^3 are computed on collapsed 7-, 5-, and 3-level versions of the scale, respectively. **Adjacent Accuracy ($\pm 1 Acc^{19}$)**: counts predictions as correct if they are either exact matches or differ by at most one level from the true label. **Average Distance (Dist)**: Also referred to as Mean Absolute Error (MAE), it captures the mean absolute difference between predicted and reference labels.

5 Results

Our team’s runs (LIS in (Elmadani et al., 2025a)), were evaluated on Codabench ⁸. Table 2 shows results on the blind test set, using the QWK and Acc^{19} metrics (Section 4.3). Initial experiments with MLP classifiers on embeddings demonstrated the importance of domain-specific pretraining. The general-purpose CAMELBERTMix_MLP performed poorly, while CAMELBERTWCE_MLP utilizing embeddings from a readability focused model demonstrated marked improvement, confirming the efficacy of task-specific pretraining. Introducing curriculum learning (P_CAMELBERTWCE_MLP) provided only a marginal gain, indicating that while progressive binning can stabilize training, its impact is limited with a frozen encoder. In contrast, curriculum learning proved more beneficial in the fine-tuning setting. PFT_CAMELBERTWCE, which progressively fine-tunes the encoder through increasingly granular label spaces, outperformed both MLP-based models. Finally, direct fine-tuning without curriculum (FT_CAMELBERTWCE) achieved the best overall QWK (82.4) and second-best Acc^{19} (57.5), slightly surpassing the progressive strategy.

⁶BAREC shared task: <https://barec.camel-lab.com/sharedtask2025>

⁷Evaluation metrics: https://github.com/CAMEL-Lab/barec_analyzer/tree/main

⁸Codabench: <https://www.codabench.org/>

To better understand model behavior, we conducted an error analysis of the two fine-tuning approaches (PFT_CAMELBERTWCE and FT_CAMELBERTWCE) on the preliminary test set. We reported conditional error rates by readability level, calculated as the percentage of incorrect predictions within each readability level. Additionally, we provided a confusion matrix in Figures 3 and 4 (Appendix A) for both the best-performing model (FT_CAMELBERTWCE) and second-best model (PFT_CAMELBERTWCE) to illustrate prediction tendencies and error patterns in greater detail. While FT_CAMELBERTWCE achieved superior overall performance, the error rate analysis (Figure 2, Appendix A) reveals that PFT_CAMELBERTWCE demonstrates lower error rates for specific readability levels (3-6, 9, 11, 12, 16, 17), suggesting that curriculum learning provides targeted improvements for certain readability levels despite lower aggregate performance. This level-specific analysis complemented the aggregate metrics by revealing where each model struggled most in distinguishing between readability levels, providing insights into the model’s systematic biases and failure modes.

6 Conclusion

We evaluated neural approaches for Arabic sentence readability assessment, comparing MLP classifiers using CamelBERT embeddings with transformer fine-tuning methods. Task-specific pretraining proved to be essential, general embeddings failed while readability-focused ones improved performance substantially. Curriculum learning provided marginal gains with frozen encoders but helped stabilize fine-tuning. Direct CamelBERT fine-tuning (FT_CAMELBERTWCE) achieved best results (QWK = 82.4, $Acc = 57.5$), surpassing baselines and slightly outperforming progressive fine-tuning. Our experiments highlight three key insights. First, task-specific pretraining is crucial for Arabic readability assessment, with domain-aligned representations significantly outperforming general-purpose embeddings. Second, curriculum learning offers modest but situational benefits. Third, direct fine-tuning remains both efficient and effective, achieving state-of-the-art performance without complex training strategies. Error analysis revealed systematic biases across difficulty levels, as illustrated in Figure 2 (Appendix A). Future work will explore hybrid architectures combining transform-

Model	Description	QWK	Acc ¹⁹
Baseline	Competition baseline	81.5	58.1
CAMeLBERTMix_MLP	Word Embedding + MLP	41.2	21.2
CAMeLBERTWCE_MLP	Word Embedding + MLP	80.5	55.7
P_CAMeLBERTWCE_MLP	Word Embedding + Progressive training of MLP	80.7	55.9
PFT_CAMeLBERTWCE	Progressive Fine-tuning	<u>82.0</u>	56.7
FT_CAMeLBERTWCE	Standard Fine-tuning	82.4	<u>57.5</u>

Table 2: Model performance on the blind test measured using Quadratic Weighted Kappa (QWK) and Accuracy (Acc¹⁹). P = progressive training strategy; PFT = progressive fine-tuning; FT = standard fine-tuning; MLP = multi-layer perceptron. CAMeLBERTWCE = CAMeLBERT-Word-CE; CAMeLBERTMix = CAMeLBERT-Mix. **Bold** indicates the best result; underlined indicates the second-best.

ers with linguistic features and multi-agent frameworks.

Limitations

This study was limited to transformer-based approaches and word embeddings. Incorporating explicit linguistic features (such as syntactic complexity, lexical diversity, and discourse markers) could complement these neural representations and potentially improve both readability prediction accuracy and model explainability.

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A Model performance analysis

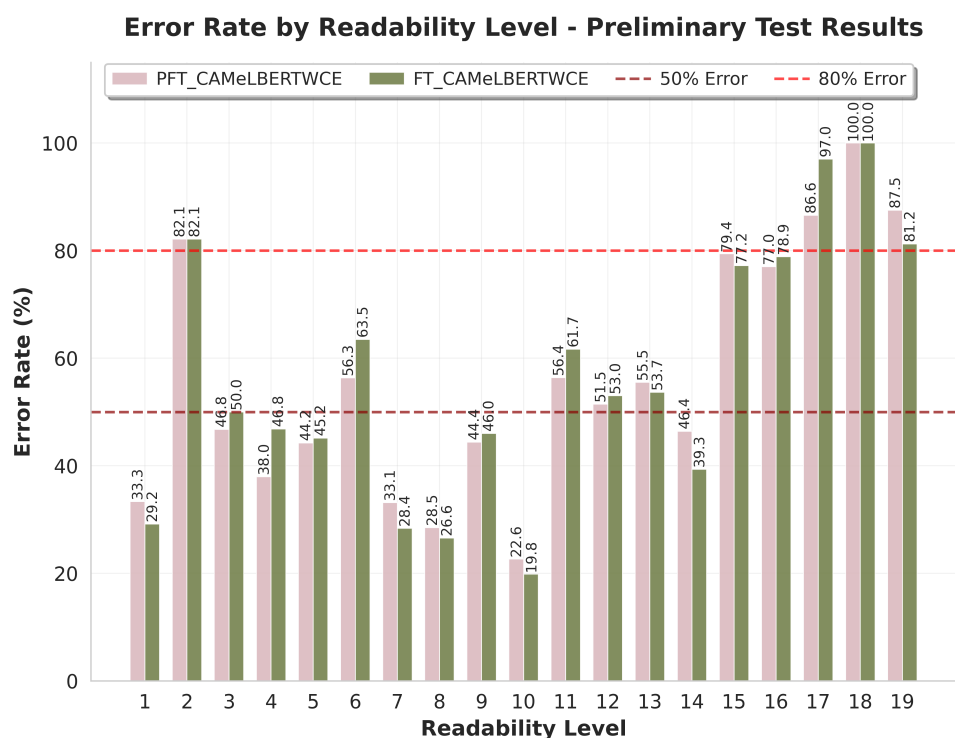


Figure 2: Error Rate by readability level on the preliminary test set

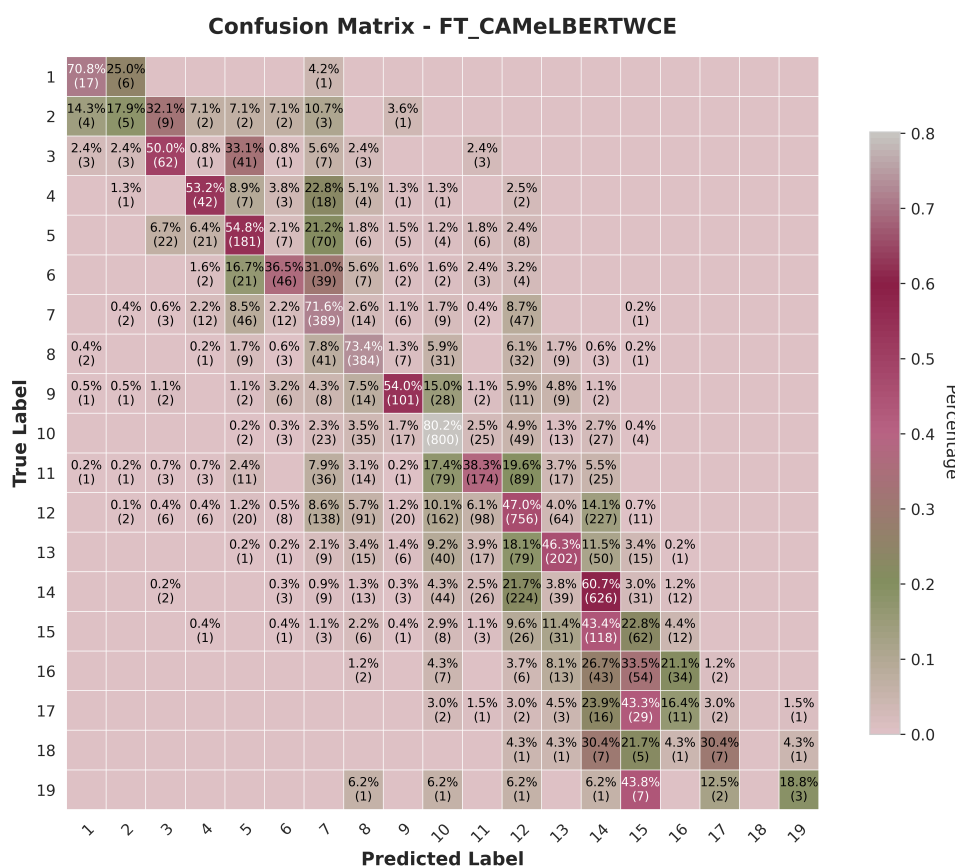


Figure 3: Prediction of FT_CAMELBERTWCE on the preliminary test set

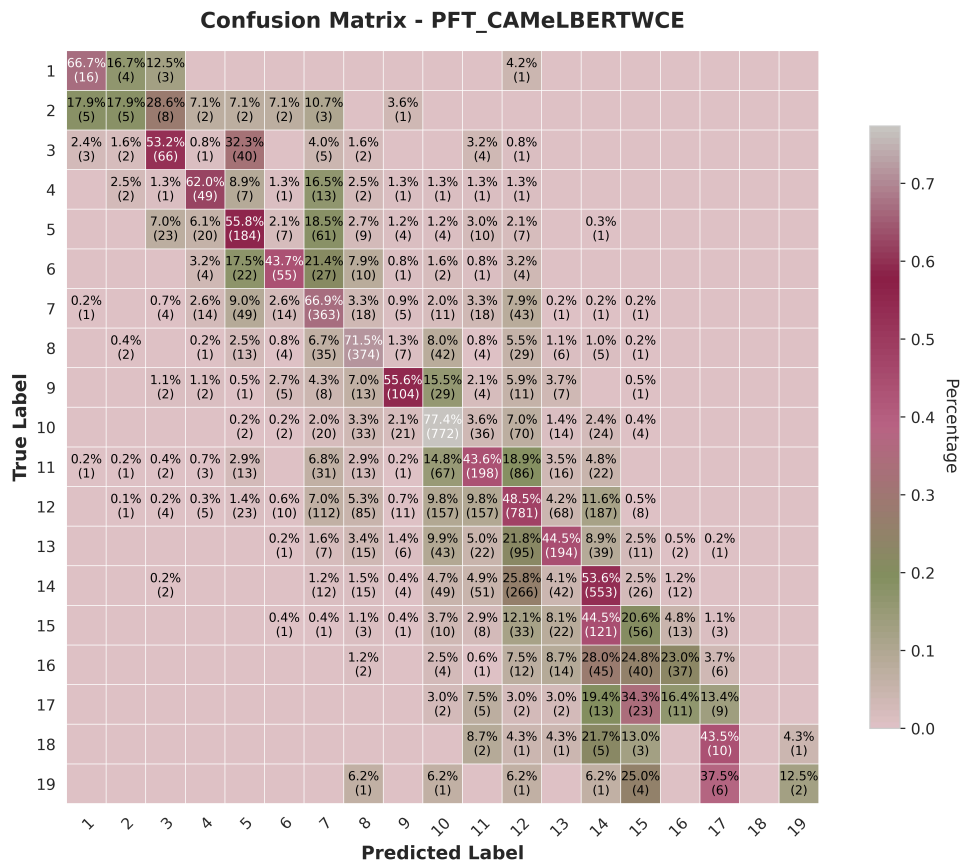


Figure 4: Prediction of PFT_CAMELBERTWCE on the preliminary test set